

Exploiting Crowd-Compression Bias in a Proper-Scoring-Rule Prediction Tournament: A Statistical Forecasting System for the 2026 FIFA World Cup

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Abstract

This paper documents a solo research campaign conducted within the Jump Probability Cup (JTC), a 2026 FIFA World Cup prediction tournament on *play.sportspredict.com* scored by Relative Brier Points (RBP)—a metric that rewards a submitted probability when it outperforms the platform’s crowd consensus on the same question and penalises it when it does not.

I built a complete statistical forecasting pipeline: point-in-time Elo ratings over 49,473 historical international matches (1872–present), a Poisson goal-rate regression with a Dixon–Coles low-score correction and Negative-Binomial overdispersion correction, a parallel ordered-logit match-result model following Hvattum and Arntzen (2010), and a StatsBomb event-level panel spanning the 2018 and 2022 World Cups. The methodology is grounded in a systematic review of thirteen sources spanning goal-based modelling, Elo-based outcome regression, small-sample model-selection theory, and wisdom-of-crowds aggregation.

Two findings anchor the paper. The first is operational: the JTC crowd consensus is structurally compressed toward 50% relative to a calibrated estimate, characterised by $\hat{p}_{\text{crowd}} \approx 0.504 + 0.61 \times (\hat{p}_{\text{mine}} - 0.5)$, fit on 772 submitted questions across the full campaign ($r = 0.83$, 95% CI on slope $[0.58, 0.64]$)—a relationship that holds essentially unchanged from an earlier fit at less than half the sample. The second is that four independent tests—a Platt-scaling recalibration, a logistic-regression/gradient-boosting blend evaluated under two out-of-sample validation schemes, a linear regression of RBP against crowd-deviation, and a closed-form result on the expected cost of pushing past a calibrated estimate under this scoring rule—all reach the same conclusion: at this campaign’s sample size, a learned model of the crowd’s own errors is not trustworthy, and every probability submitted in this campaign came from the hand-built probabilistic pipeline instead.

Across 86 settled matches and 935 individually scored questions as of this writing, the campaign holds a cumulative total of approximately +3,415 RBP at a 70% crowd-beat rate, moving from the top 5.6% to the top 1.1% of an open field of nearly 3,900 registered participants over the final three weeks of the group and knockout stages. I report the campaign’s best and worst individual results in full, including an -80.83 RBP loss that independently confirms a previously identified, still-active failure mode, rather than smoothing over the record.

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1 Introduction

1.1 Motivation and Problem Framing

The 2026 FIFA World Cup, hosted across the United States, Canada, and Mexico in June and July 2026, poses an unusual forecasting problem. The object of prediction is not a continuous outcome—goals, points, margins—but a portfolio of discrete binary and three-way questions resolved at match level: which team wins, whether each team scores, whether a named player records a shot on target, how many corners are taken in the first half. This structure is closer to the question sets on platforms such as Good Judgment Open and Metaculus than to a standard sports-betting market.

This paper documents an ongoing research campaign I have run inside a specific competitive environment: the Jump Probability Cup (JTC), a World Cup prediction tournament on `play.sportspredict.com` scored by *Relative Brier Points* (RBP). Under the RBP mechanism, a submitted probability earns positive points when its squared error is lower than the platform-revealed crowd consensus on the same question and negative points otherwise. RBP is therefore a *relative* proper scoring rule: the optimal strategy remains reporting my true belief, yet the metric rewards calibration relative to the crowd rather than relative to an absolute ground truth. The competitive advantage lies not in outperforming the crowd on raw accuracy but in identifying the systematic ways the crowd’s aggregate deviates from a calibrated probability.

Four contributions structure this work. First, I present a complete statistical forecasting pipeline—Elo ratings, a Poisson goal-rate regression, a Dixon–Coles correction, Negative-Binomial overdispersion, an ordered logit, and a StatsBomb event-level panel—grounded in a systematic literature review and validated by walk-forward backtesting against 49,000-plus international matches. Second, I fit and periodically revalidate an empirical crowd-bias correction characterising the JTC crowd’s systematic compression toward 50%. Third, I report four independent tests of whether a learned model of my own estimate and the crowd’s should replace the hand-built pipeline, all four of which answer no, and I show why that answer is expected rather than surprising given the campaign’s sample size. Fourth, I document a live, resolved track record across 935 questions spanning the group stage through the quarterfinal, categorise outcomes by pricing technique, and trace the root causes of both the largest gains and losses without smoothing over the ones that went badly.

1.2 Scope of This Document

All model code, data files, and per-match research records referenced in this paper reside in a private git repository (`sportspredict_research/`). File paths cited throughout (e.g., `model/elo.py`, `datasets/master_question_dataset.csv`) refer to paths within that repository. The master question dataset, at its most recent rebuild, contains 943 rows across 85 source match files; the most recent settled match, France versus Morocco (quarterfinal, 2026-07-09), is not yet folded into that rebuild, and its verified standalone total is reported alongside the master-dataset figure rather than silently merged into it (§8). Every number in this paper traces to a specific file read at the time of writing; where two documents in the repository disagree because they were computed at different points in a still-running campaign, both figures are reported with the disagreement stated plainly rather than resolved by assumption.

2 Literature Review

The modelling choices in this project rest on a systematic review of thirteen sources, organised by modelling lineage rather than by publication date.

2.1 Goal-Based and Poisson-Family Models

Dixon & Coles (1997) (Dixon and Coles, 1997) is the foundational contribution in this lineage. It treats home and away goals as two independent Poisson distributions:

$$X \sim \text{Poisson}(\lambda), \quad \lambda = \alpha_i \cdot \beta_j \cdot \gamma$$

$$Y \sim \text{Poisson}(\mu), \quad \mu = \alpha_j \cdot \beta_i$$

where α_i is the attack strength of team i , β_i is the defensive weakness, and γ is a home-advantage multiplier. In log-linear form this is a Poisson GLM with team dummy variables—attack and defence—plus a home-effect term. Two enduring contributions characterise this work: the *tau correction*, a multiplicative adjustment to the four low-score cells (0–0, 1–0, 0–1, 1–1) that repairs independent Poisson’s tendency to under-predict draws:

$$\tau_{\lambda,\mu}(x, y) = \begin{cases} 1 - \lambda\mu\rho & \text{if } x = 0, y = 0 \\ 1 + \lambda\rho & \text{if } x = 0, y = 1 \\ 1 + \mu\rho & \text{if } x = 1, y = 0 \\ 1 - \rho & \text{if } x = 1, y = 1 \\ 1 & \text{otherwise} \end{cases}$$

where ρ is a single dependence parameter estimated from data, and exponential time-decay weighting, $\phi(t) = e^{-\xi t}$.

Karlis & Ntzoufras (2003) (Karlis and Ntzoufras, 2003) model the home/away goal correlation directly via a bivariate Poisson (BP) distribution constructed from three independent latent Poisson components. If Z_1, Z_2, Z_3 are independent Poisson with means $\lambda_1, \lambda_2, \lambda_3$, then $(X_1, X_2) = (Z_1 + Z_3, Z_2 + Z_3) \sim \text{BP}(\lambda_1, \lambda_2, \lambda_3)$, with $\text{Cov}(X_1, X_2) = \lambda_3 \geq 0$. The base BP model can only capture positive correlation, a structural constraint relevant to international football, where the empirical data—particularly at the international level—

frequently exhibits negative goal correlation, motivating the Sarmanov-family extension below.

Michels, Ötting & Karlis (2023) (Michels et al., 2023) reframe the Dixon–Coles tau correction as a special case of the Sarmanov family of bivariate distributions and demonstrate empirically—motivated by women’s football, where negative goal correlation is as strong as $\rho \approx -0.25$ to -0.40 —that the DC correction achieves at most $\rho \approx -0.05$ to -0.08 at typical scoring rates. This finding directly bounds the benefit I could expect from re-fitting the ρ parameter in this project (§4).

Crowder, Dixon, Ledford & Robinson (2002) (Crowder et al., 2002) extend the DC framework by allowing attack and defence parameters to evolve as a state-space process rather than remaining fixed within a season. A full implementation of this formulation is beyond this project’s scope; I substitute DC’s exponential time-decay combined with periodic re-fitting, which approximates time-varying strength without a full particle filter.

2.2 Rating-Based Models and Outcome Regression

Hvattum & Arntzen (2010) (Hvattum and Arntzen, 2010) furnish the direct template for half of my modelling stack. The standard Elo update,

$$R'_h = R_h + K \cdot (S_h - E_h), \quad E_h = \frac{1}{1 + 10^{(R_a - R_h)/400}}$$

together with a goals-based variant in which the K -factor scales with margin of victory, feeds a *proportional-odds ordered logistic regression* that maps the Elo difference $x = R_h - R_a$ directly to Home/Draw/Away probabilities:

$$P(\text{Away win}) = \sigma(\kappa_1 - \beta x), \quad P(\text{Draw}) = \sigma(\kappa_2 - \beta x) - \sigma(\kappa_1 - \beta x)$$

$$P(\text{Home win}) = 1 - \sigma(\kappa_2 - \beta x)$$

where $\sigma(\cdot)$ is the logistic function and $\kappa_1 < \kappa_2$ are fitted cutpoints. Applied to 14 seasons and 30,524 English league matches, this approach outperforms uniform and historical-frequency baselines but falls short of bookmaker-implied probabilities—a result that directly motivates treating market prices as a primary rather than secondary input wherever they are available.

2.3 Scoring-Rule Verification and Calibration

Constantinou & Fenton (2012) (Constantinou and Fenton, 2012) introduce the Ranked Probability Score (RPS) as a scoring rule that respects football’s ordinal outcome structure:

$$\text{RPS} = \frac{1}{2} [(p_A - e_A)^2 + (p_A + p_D - e_A - e_D)^2]$$

where outcomes are ordered Away < Draw < Home on the goal-difference axis. Five canonical benchmark scenarios demonstrate that the plain Brier score—treating the three outcomes as nominal—can rank a forecaster who is confidently wrong in direction no worse than one who is one category off, a pathology RPS avoids by construction. Whether the JTC platform’s weighted-Brier formula inherits this pathology remains an open empirical question (§10).

Murphy (1973) (Murphy, 1973) supplies the foundational calibration–refinement decomposition of the Brier score,

$$\mathbb{E}[S(P, X)] = \underbrace{\text{Var}(X)}_{\text{UNC}} - \underbrace{\mathbb{E}_P\{[\mathbb{E}(X|P) - \mathbb{E}(X)]^2\}}_{\text{RES}} + \underbrace{\mathbb{E}_P\{[\mathbb{E}(X|P) - P]^2\}}_{\text{REL}}$$

where RES is resolution (discriminative power) and REL is reliability (calibration error). This decomposition frames §7: it distinguishes “I cannot separate strong from weak teams” (low RES) from “my confidence is systematically miscalibrated” (high REL), two failure modes requiring different remedies.

Foulley (2021) (Foulley, 2021), applied to four seasons of UEFA Champions League group-stage data, yields three findings that bear directly on this project: a practical estimation procedure for REL, RES, and UNC using isotonic regression, which my backtesting diagnostics adapt; the result that Draw is structurally the hardest outcome to discriminate (Harrell’s $C \approx 0.62$ for Draw versus 0.79–0.82 for Home/Away); and a closed-form derivation showing that a Poisson log-linear model with a shared Elo-difference covariate of opposite sign in each team’s predictor automatically induces negative home/away goal correlation—justification for incorporating the Elo difference directly into the Poisson means rather than relying solely on the tau correction.

2.4 Small-Sample Model Selection and Trading-Rule Replication

A fourth thread, absent from the literature review’s original scope but load-bearing for §6, concerns when a learned estimator should be trusted at all. Peduzzi et al. (1996) and its extension in Riley et al. (2019) establish the events-per-variable (EPV) threshold—roughly 10 to 20 settled outcomes per fitted parameter—below which logistic-regression coefficients are provably unstable, a threshold this campaign’s own question-level sample sits well under for any multi-feature blending model. Harvey et al. (2016) documents that a majority of published trading-rule discoveries fail to replicate out of sample, the direct analogue of the named situational rules in §5, several of which have already required evidence-forced revision within this campaign’s own short history. Merkle et al. (2017) supplies the Brier-score decomposition appropriate to a heterogeneous question portfolio, used in §8. Mamlouk et al. (2024) shows empirically, in a sports-betting context, that model selection by calibration rather than raw accuracy is the decision that actually preserves return, the reason Brier score, not classification accuracy, is the only metric used throughout §6.

2.5 Wisdom of Crowds and Probability Aggregation

Lichtendahl, Winkler & Bickel (2020) (Lichtendahl et al., 2020) analyse the 2018 FIFA World Cup public forecasting contest (fifaexperts.com), scored by Brier score. Simple un-weighted averaging of participants’ probabilities and betting-market consensus both ranked at or near the top of the leaderboard, outperforming most individual participants; more sophisticated weighting schemes—including Budescu–Chen regret-minimisation (Budescu and Chen, 2015) and online expert-advice algorithms—did not reliably outperform simple averaging. The paper also proves that the minimax-optimal forecast under Brier scoring with zero information is the uniform forecast $(1/3, 1/3, 1/3)$, formal justification for the standing rule (§5) against submitting 0% or 100% under any circumstances.

This project’s core finding—that the JTC’s free-entry, financially undisciplined crowd constitutes a structurally compressed version of a calibrated probability—is consistent with the favourite–longshot bias literature (Snowberg and Wolfers, 2010): participants without financial exposure systematically avoid extreme probabilities, causing the aggregate to regress toward 50/50.

3 Data

3.1 Source Inventory and Reliability Tiers

The most distinctive data artefact in this project is a tiered source-reliability audit, derived empirically from settled matches spanning the complete group stage and the knockout phase through the quarterfinal, encompassing over 900 individually graded questions—not asserted a priori, but constructed by tracing specific wins and losses to the data source driving each decision. Following Gebru et al. (2021)’s Datasheets-for-Datasets framework, each tier is documented with its intended use, the basis for its assigned trust level, and known failure modes.

Table 1: Data source reliability tiers

Source	Use	Trust basis	Failure mode
Tier 1 — reliably profitable; weight heavily			
ESPN roster API	Per-player SOT, fouls, offsides from prior matches (“GD1” data)	Largest single wins: +22, +25, +20, +44 RBP on offsides/SOT propositions	Blowout-reversal: team shape abandoned when chasing 3+ goals
Main sportsbook consensus (FanDuel + DraftKings + Betfair + bet365 agreeing)	Match winner, BTTS, major propositions; player <i>absence</i> from scorer markets is itself informative ($p < 0.35$)	When ≥ 3 major books agree within 5 pp, near-ground-truth	Following a niche aggregator disagreeing with main-book absence: -11.03 RBP (Ramos, POR–UZB)
FootyMetrics referee career statistics	Card and penalty questions	Career sample ≥ 80 matches per referee	Requires blowout discount of -30 to -40% in extreme mismatches

Continued on next page

Table 1 continued

Source	Use	Trust basis	Failure mode
Smarkets exchange (liquid: bid+offer)	Match winner, BTTS, OU goals, OU corners, OU SOT	Financially-incentivised, two-sided; mid-price is arbitrage-free	Illiquid (offer-only) markets require a 5.5% haircut
Polymarket (>\$1M traded)	Match-winner probability	High liquidity, financially incentivised	Below \$500K volume, a secondary signal only
Tier 2 — useful with specific adjustments; never a primary driver in isolation			
Footystats 10-match foul averages	Foul-count magnitude prior	Reliable for magnitude, not for direction	Requires predicting which team will be defending
Model output (Elo / Poisson / ordered logit)	Match structure baseline	Calibrated against 49K+ historical matches	Weight at $\leq 10\%$ once Elo gap > 350 points; 90% weight to market
Tier 3 — avoid or discount heavily			
Niche aggregator scorer odds (when player absent from main books)	—	—	Goal.com/Ramos failure: -11.03 RBP
Personal conviction overriding a researched figure	—	—	The worst-ever individual result before the campaign's largest single-question loss (§8) shared this root cause

The operative source hierarchy, applied whenever sources conflict: verified ESPN GD1 in-

dividual behavioural data (with a hard floor of 0.25 on any count-statistic question) > main sportsbook consensus > referee career statistics > high-volume Smarkets liquid market > club-level SOT data > Polymarket (above \$1M) > Tier-2 sources > model output in isolation.

3.2 Data Pipeline Architecture

The pipeline follows a staging $\rightarrow$ intermediate $\rightarrow$ mart architecture in the tradition described by Gebru et al. (2021) and visible in open football-data pipelines such as `dcaribou/transfermarkt-datasets`. Raw acquisition is immutable: `bot/fetch_markets.py` retrieves all open matches and markets from the SportsPredict v1 API and writes them immediately to disk in JSONL format, one line per match, without memory buffering prior to write, a convention I formalised after two early data-loss incidents. A reverse-engineering exercise documented in `bot/API_EXPLORATION_2026-06-27.md` identified two distinct API namespaces: a documented external namespace (`api.sportspredict.com/api/v1`, with two functional routes, `/matches` and `/markets`) and an undocumented browser-only namespace (`api.sportspredict.com/api`, cookie-authenticated) that additionally exposes settled-trade history, leaderboard rank, and per-user score summaries.

Classification and cleaning follow: `bot/classify_markets.py` categorises every market question as model-coverable (`match_winner`, `total_goals_under/over`, `btts`) or `no_model` (fouls, cards, corners, offsides, player propositions), with the latter priced via the manual research and rules workflow. A live Smarkets event-discovery step employs `limit=300` plus client-side string filtering, since the `name=` API query parameter is confirmed non-functional across all matches—it silently returns an unfiltered page with HTTP 200, a non-obvious failure mode now standard to check for.

Master dataset construction is the final stage: `datasets/build_master_dataset.py` reads all per-match files and produces one flat row per (match, question) in `datasets/master_question_dataset.csv` the authoritative derived dataset. Raw files are never edited or overwritten; all reprocessing reads from the original and writes to a new derived output.

3.3 Master Dataset Documentation

At its most recent rebuild, the master question dataset contains 943 rows across 85 source match files from `data/external_markets/` and `matches/`, of which 921 carry a settled RBP value and 943 a recorded outcome. Full column documentation resides in `datasets/MASTER_DATASET_DICTIONARY.md`.

Question-number join keys are unreliable as a matching mechanism: platform numbering (Q1, Q2, etc.) drifts between the draft submission screen and the final scoring record, verified concretely in two files (`eng_cro_2026-06-17.json`: q5 pre-match versus Q7 post-match for the same question; `cro_pan_2026-06-23.json`: a one-position rotation). The build script resolves this via topic-word tokenisation and Jaccard similarity matching (≥ 0.35 , requiring ≥ 2 shared tokens), never by assuming positional consistency, and the core scored fields are unaffected by the problem since they derive from the post-match record’s own internally consistent numbering. Pre-match research enrichment covers a genuine minority of rows—roughly a quarter carry a machine-readable `recommended_estimate` from before the match, concentrated in matches after mid-June, a historical data gap rather than a matching artefact. A small, explicitly flagged set of questions across the dataset show a submitted value differing from the researched recommendation by more than 0.03, including the documented ENG–CRO Kane transcription error discussed in §8. Squad value, age, and FIFA ranking enrichment is null for five teams (Cape Verde, Curaçao, DR Congo, Haiti, Ivory Coast), absent from the source Transfermarkt extract rather than a join failure.

3.4 A StatsBomb Event-Level Panel

A newly constructed StatsBomb open-data panel, restricted to the 2018 and 2022 World Cups—the subset of the broader 80-competition, roughly 3,961-match public corpus with schema-identical, fully validated event coverage—gives 128 matches, flattened into a 6,131-row player-match panel and a 257-row team-match panel. I validated this panel before using it for anything: total goals reproduce known tournament history exactly (169 goals in 2018, 172 in 2022, 2.641 and 2.688 per match respectively), and top scorers reproduce correctly (Mbappé’s eight goals across the 2022 tournament, including the final, appear row-for-row).

The panel’s value was tested directly rather than assumed. Against a settled match, Portugal versus Croatia, the pure StatsBomb-history model produced +93.01 RBP-equivalent

across fourteen comparable questions, while the actual submission for that match—StatsBomb history blended with live market and tournament-form evidence—scored +120.30 RBP, and the actual submission’s advantage held despite the pure-history model individually winning six of the fourteen questions outright. Every StatsBomb-derived rate used in this campaign is accordingly treated as a prior blended with the current tournament’s own live form, never a standalone replacement for it.

3.5 Measurement Divergence Across Data Sources

A finding worth stating plainly rather than absorbing silently into the pipeline: ESPN and StatsBomb do not count the same underlying events identically. Across matches covered by both sources, ESPN records roughly 20% fewer fouls, 27% fewer yellow cards, 38% more shots on target, and 30% more corners than StatsBomb for the same matches, while total-shot counts are close to parity between the two. The practical consequence is that raw counts must never be pooled across a `data_source` column without a source-specific correction factor; every count-based feature used anywhere in this pipeline is tagged by its source and never silently averaged with a count from the other source.

4 Methodology

I run two parallel approaches to the core match-result question—a goals-based pipeline and a direct outcome-regression pipeline—following the posture recommended by Hvattum and Arntzen (2010). For proposition markets outside the reach of either model (fouls, cards, offsides, individual player propositions), a live Smarkets market price is the primary input, supplemented by the rules framework of §5.

4.1 Point-in-Time Elo Ratings (`model/elo.py`)

Every national team carries a single numeric rating updated after each historical match using the standard Elo update rule. Three domain-specific refinements are applied: a goal-difference multiplier, $K_{\text{eff}} = K_0 \cdot \frac{11+|\Delta g|}{8}$, which imposes diminishing returns to very large victories; a home-advantage offset of a flat +100 rating points applied to the home team when the match is not on a neutral venue; and a match-importance K -factor, $K = 60$ for World Cup knockout matches, decreasing to $K = 20$ for friendlies. Ratings are computed over the full `martj42` international results panel (49,473 matches, 1872–present), strictly point-in-time with no lookahead. The initial rating $R_0 = 1500$ applies at a team’s first appearance; by 2026 this matters only for recently FIFA-affiliated nations with short competitive histories.

A data defect that silently froze every in-progress rating. Pulling Elo differentials for five knockout-stage matches in early July surfaced a bug affecting every rating used in this paper up to that point. Argentina’s pre-match rating is identical—2189.5282—across all three of its group-stage rows in the project’s Elo panel, despite winning those matches 3–0, 2–0, and 3–1. The cause traces to `data/international_results/results.csv`, the historical panel this project treats as ground truth: it records every 2026 World Cup match’s score as NA, a fixture list rather than a results file, so `model/elo.py` has never processed a single real 2026 result. Every “current” rating used anywhere in this campaign has actually been frozen at pre-tournament, early-June values. This is not a lookahead problem—nothing from the future leaked into an earlier estimate—but its opposite: a model made needlessly uninformative by failing to update on results that were already known. I built a point-in-time replay script (`ml/backtests/r16_point_in_time_elo_replay.py`) that reconstructs each team’s actual group-stage and knockout results through the exact Elo formula above,

with true home/neutral status confirmed from venue data rather than assumed. Applied to five retrospective matches, the replay moved Elo differentials by 12 to 35 points relative to the frozen baseline (Table 2). The workaround is match-by-match, not a pipeline-level fix: it needs to be re-run for every future knockout fixture until the underlying source file, not just individual matches’ numbers, is corrected.

Table 2: Corrected point-in-time Elo differentials versus the frozen, pre-tournament baseline, five retrospective matches.

Matchup	Corrected	Frozen (stale)	Difference
Switzerland — Colombia	−79.45	−111.92	+32.47
Argentina — Egypt	+363.62	+379.78	−16.16
United States — Belgium	−129.52	−117.22	−12.30
Portugal — Spain	−142.62	−169.05	+26.44
Mexico — England	−70.36	−105.58	+35.22

4.2 Poisson Goal-Rate Regression (`model/poisson_goals.py`)

The Elo difference between two teams is translated into expected goal counts via a Poisson GLM:

$$\log(\lambda_{\text{home}}) = b_0 + b_1 \cdot \mathbf{1}[\text{home}] + b_2 \cdot \Delta R, \quad \log(\lambda_{\text{away}}) = b_0 + b_2 \cdot (-\Delta R)$$

where $\Delta R = R_{\text{home}} - R_{\text{away}}$ is the pre-match Elo difference and $\mathbf{1}[\text{home}]$ is the home-advantage indicator, suppressed to zero for neutral-venue matches. Each historical match contributes two observations, one per team, exponentially down-weighted by recency with a half-life of approximately 2.4 years. Fitted on $n = 49,400$ matches ($n_{\text{obs}} = 98,800$ team-match observations) with exponential recency weighting (decay constant $\xi = 0.0008 \text{ day}^{-1}$), Table 3 reports the MLE estimates.

Table 3: Fitted Poisson GLM parameters

Symbol	Parameter	Estimate	SE	Interpretation
b_0	Intercept	0.1041	0.0141	Log-goals at equal Elo, neutral venue
b_1	Home advantage	0.2302	0.0211	Log-goals uplift at home ($e^{b_1} \approx 1.26 \times$ goals)
b_2	Elo difference	0.001810	3.68×10^{-5}	Log-goals uplift per Elo point of advantage
ξ	Decay constant	0.0008 day^{-1}	—	Exponential recency weight; half-life $\approx 2.4 \text{ yr}$

A second, independent bug was found and resolved in `build_design_matrix()`: the training code set the home-advantage indicator without consulting the match’s `neutral` flag, while every downstream file (prediction, backtesting, rho-fitting) correctly conditions on it, a train/serve mismatch affecting roughly 26% of the historical panel’s rows. The fix has been applied and the model re-fitted; the coefficients in Table~3 reflect the corrected model. Worth recording directly rather than smoothing over: I had predicted the bug was attenuating b_1 downward from a rough prior of ~ 0.35 , and the opposite turned out to be true—the buggy coefficient (0.256) was higher than the fixed one (0.230), because neutral-venue “home” teams skew slightly toward the higher-rated side by labelling convention, so including them inflated the home-advantage signal rather than suppressing it. The fix is correct regardless of which direction the bug ran; the remaining gap between $b_1 = 0.230$ and the ~ 0.35 prior is better explained by attenuation bias from a noisy Elo covariate (§7) than by this bug.

4.3 Dixon–Coles Correction and Negative-Binomial Overdispersion

The Poisson means above are used to construct a full scoreline probability grid. Two corrections are applied sequentially. The Dixon–Coles tau correction’s dependence parameter is estimated by maximum-likelihood grid search over the historical panel; the current fit, $\hat{\rho} = -0.05$, is close to the theoretical maximum Michels et al. (2023) finds achievable at typical scoring rates, so re-fitting via a full bivariate Poisson would not materially improve

calibration. A Negative-Binomial refit of the marginals addresses a separately documented overdispersion pattern in the total-goals tail: an overdispersion parameter α is fitted by MLE, with $\alpha = 0$ reducing to the Poisson special case. From the full scoreline grid, every model-coverable market is computed analytically: match result, total goals over/under, both-teams-to-score, and “team scores $\geq N$.” Following Foulley (2021)’s Appendix B derivation, incorporating the Elo difference directly into the Poisson means automatically induces negative home/away goal correlation without a separate bivariate-Poisson λ_3 term, theoretical grounding for the current model structure.

Table 4: Fitted Dixon–Coles and Negative-Binomial parameters ($n_{\text{obs}} = 98,800$)

Symbol	Parameter	Estimate	SE	Interpretation
$\hat{\rho}$	DC low-score dependence	−0.05	—	Grid-search MLE; near theoretical achievable maximum (Michels et al., 2023)
$\hat{\alpha}$	NB dispersion	0.0992	0.0131	Variance inflation: $\text{Var}(X) = \lambda + \hat{\alpha}\lambda^2$

4.4 Ordered Logit (`model/ordered_logit.py`)

The direct implementation of Hvattum and Arntzen (2010)’s approach fits a proportional-odds ordered logistic regression against the three-way match outcome using Elo difference and a home-advantage indicator as covariates, plus two cutpoints. Because it is calibrated against the exact quantity it predicts, this model backtests with superior outcome-level calibration relative to the goals-based pipeline in the high-confidence range. It serves as the primary instrument for match-result markets; the goals-based pipeline handles all goal-total, BTTS, and proposition questions.

Table 5: Fitted ordered logit parameters

Symbol	Parameter	Estimate	SE	Interpretation
β	Elo slope	0.005199	0.000176	Log-odds increase per Elo point of advantage
δ	Home advantage	0.3771	0.0745	Log-odds boost at non-neutral venue
κ_1	Lower cutpoint	-0.7702	0.0643	Away-win / Draw boundary
κ_2	Upper cutpoint	0.5549	0.0634	Draw / Home-win boundary

At equal Elo on a neutral venue, the implied probabilities are $P(\text{Away}) = \sigma(\kappa_1) = 0.316$, $P(\text{Draw}) = 0.319$, $P(\text{Home}) = 0.365$.

4.5 Live Market Pricing via Lambda-Fitting

For questions outside both models’ scope, Smarkets market prices serve as the primary input. A Poisson lambda is recovered from each available over/under threshold via root-finding,

$$\hat{\lambda} = \text{brentq}(f(\lambda) = \text{Poisson.sf}(k, \lambda) - p, (0.001, 100))$$

where p is the de-vigged market over-probability and k is the threshold. Multiple thresholds for the same underlying quantity give internal consistency checks: if two over/under markets on the same statistic imply $\hat{\lambda}$ values differing by more than ~ 0.05 , I flag the model as internally inconsistent before submission. For head-to-head comparison questions, the Skellam distribution yields the exact analytic probability (Skellam, 1946): if $X \sim \text{Pois}(\lambda_A)$ and $Y \sim \text{Pois}(\lambda_B)$ independently, $P(A > B) = 1 - F_{\text{Skellam}}(0; \lambda_A, \lambda_B)$. For compound questions with no direct market (for example, “will both teams receive at least one card”), I decompose each side’s own independently measured rate rather than assume a shared or symmetric prior—the single best individual result in the campaign to date came from exactly this method (§8).

4.6 Validation: Expanding-Window, Time-Ordered Backtesting

All statistical models are validated via a time-ordered walk-forward backtest (`model/backtest_harness.py`), explicitly not random k -fold cross-validation, since match outcomes are temporally ordered and model coefficients are fitted using only recency-weighted prior data. The harness reports mean Brier score and mean RPS for the match-result market against a naïve $\frac{1}{3}/\frac{1}{3}/\frac{1}{3}$ baseline, together with a per-decile calibration table.

5 Crowd-Bias Model and Decision Rules

5.1 The Central Empirical Finding

I fit a direct empirical relationship between my submitted estimates and the platform-revealed crowd consensus across all 772 submitted questions in the campaign with a paired crowd estimate, spanning the group stage through the quarterfinal:

$$\hat{p}_{\text{crowd}} \approx 0.504 + 0.61 \times (\hat{p}_{\text{mine}} - 0.5) \quad (1)$$

Full regression statistics: $r = 0.829$, $R^2 = 0.688$, residual SD = 7.19 pp, 95% CI on slope = [0.580, 0.638]. A time-ordered out-of-sample check (first 540 questions chronologically as training, remaining 232 as test) gives RMSE = 5.85 pp and mean bias = 1.08 pp.

This relationship is worth reporting alongside an earlier fit rather than only in its final form: at $n = 384$, restricted to the group stage alone, the same regression gave slope 0.60, intercept 0.511, and out-of-sample RMSE 8.44 pp. Doubling the sample into the knockout rounds moved the slope by 0.01 and the intercept by 0.007, and tightened the out-of-sample RMSE rather than degrading it. This is the pattern I would expect if the compression ratio is a genuine platform-level behavioural constant—the same \$3,900 registered participants, scored under the same rule, with no financial exposure, generate crowd estimates under an unchanged incentive structure whether the match is a group-stage fixture or a quarterfinal—rather than an artefact of an early, unrepresentative sample.

The 0.61 slope indicates the crowd compresses any submitted estimate by roughly 39% of the distance back toward 50%, with a small uniform upward anchor of half a percentage point. Decomposing by direction: when my estimate falls below 0.5 (500 of 772 questions), my mean (0.334) sits close to the realised base rate, while the crowd’s mean (0.403) overestimates the probability of “unlikely” outcomes by roughly 6.9 pp; in the above-0.5 direction (252 questions), the crowd’s mean (0.588) underestimates “likely” outcomes by 5.1 pp relative to my own mean (0.639). In aggregate, the crowd is less extreme than my estimate on 73.2% of questions and more extreme on only 22.5%, direct evidence of systematic compression rather than simple noise around my own number.

The crowd-bias model does not generate primary estimates; it anticipates the structural gap between a planned submission and the expected crowd consensus, letting me quantify in

advance the directional magnitude of any proposed deviation. I treat it as a live, periodically revalidated empirical parameter, updated whenever a new batch of resolved questions is large enough to shift the regression coefficients by more than one standard error. A necessary caveat on the performance record: across an early 85-question calibration window, the crowd’s raw Brier score in fact exceeded my own, and my head-to-head beat-crowd rate at that sample point was only 52.9%. The large positive cumulative RBP derives from winning disproportionately on higher-leverage questions where the crowd is most wrong in magnitude, not from superior accuracy in a simple count sense—a high-variance, positive-expected-value strategy, not a uniformly reliable edge.

5.2 Named Situational Rules

Beyond the baseline crowd-compression model, fifteen named rules encode situational corrections derived by tracing specific match outcomes to their proximate cause, each carrying an explicit evidence grade.

Rule 1 (match-winner questions) sets the estimate as $\frac{1}{2}(\hat{p}_{\text{crowd consensus}} + \hat{p}_{\text{verified sharp market}})$. Across the first three settled match-winner questions, this formula was never the single worst-performing input of the three available—a minimax-regret rationale rather than an accuracy claim, given the limited sample it was built on.

Rule 7 (underdog proposition edge) states that when a team is priced below \$15% to win, the edge concentrates in propositions about the underdog, not the headline match-winner. First confirmed following Germany 7 – 1 Curaçao (+72.47\$ RBP), where more than 80% of the total edge derived from five Curaçao-specific propositions, and subsequently confirmed at $n \geq 5$.

Rule 8 (high draw probability: shrink comparison propositions) specifies that when the pre-match draw probability exceeds \$25%, any “Team A more [count statistic] than Team B” estimate is shrunk toward 0.50:

$$p_{\text{final}} = 0.65 \times p_{\text{model}} + 0.35 \times 0.50 \quad (2)$$

Confirmed on four matches with a net positive result, then refined to a 0.55–0.60 lean after a counterexample (a one-goal draw with a ten-foul differential) showed the shrinkage could

overcorrect.

Rule 12 (pre-submission risk gate) requires that before submitting a no-market comparison proposition, I remove the single most extreme game from either team’s ESPN sample and recompute the estimate. If removal shifts the estimate by more than 10 pp, I treat it as low-confidence and pull it toward the crowd-formula prediction rather than retaining full distance from 50%.

Rule 15 (extreme-underdog forward suppression) sets the primary attacking forward’s SOT and scoring probability into roughly a 0.20–0.35 band when a team is priced below \$ 15 – 20% to win. *This rule underwent a major evidence – forced revision when it failed severely* (–31.86\$ RBP) on a player whose suppressing evidence came from a match in which his team was comfortably dominant, not chasing a deficit—the opposite tactical role. The rule now branches explicitly on game state: a blowout with the team defending a lead gets suppression; a competitive scoreline with the team chasing does not, and the suppression may even invert.

A hard submission floor prohibits any submission below 0.25 on count-statistic questions, adopted after a –42.56 RBP result (Haiti 2+ offsides, submitted from a raw Poisson output near zero, resolved YES after Brazil went 3–0 up and Haiti’s defensive shape collapsed) that a 0.25 floor would have capped at roughly –10 to –15 RBP. *Rule 6* prohibits submission of 0 or 100%, formally motivated by Lichtendahl et al. (2020)’s minimax proof that the uniform forecast is minimax-optimal under Brier scoring with zero information, and empirically motivated by the worst individual result in the campaign’s early record, a submission at 1.00 on a personal-conviction override against a researched figure of 0.58–0.60 that scored the maximum possible penalty when the match drew.

A cross-match pattern that generalised and then partly failed. A synthesis across five settled matches and 75 questions identified a sharp, near-symmetric pattern in two specific question shapes. Suppressing a named player’s proposition on a genuine multi-game production drought won 7 of 7 tested instances, net +144.67 RBP; blending a favourite team’s own shots-on-target threshold upward from a thin market line toward its recent empirical average lost 5 of 5, net –155.64 RBP. I wrote this up as a standing rule—default to the market, deviate only downward, never upward, on a favourite’s own SOT threshold—and it was live-tested twice in the following weeks with opposite results. In Brazil versus Norway,

staying at the market on Brazil’s own 6-or-more-SOT question rather than blending toward the higher empirical average earned +22.57 RBP, the second-biggest win of that match, and the first live confirmation that the rule improves real decisions rather than just fitting past data. Days later, in Canada versus Morocco, the same question shape (Canada’s own 4-or-more-SOT threshold) produced the campaign’s worst single result to date: I deviated from the market toward the empirical average anyway, submitted 0.80 against a market-implied 0.577 and a crowd of 0.46, and Canada did not reach the threshold, costing −80.83 RBP. This was not a failure to know the rule; it was a live test of a theoretically motivated argument for deliberately deviating past a calibrated market estimate near a leaderboard-standing consideration (§6), applied to a signal that was not in fact high-confidence, and it lost exactly the amount the underlying theory predicts such a deviation should cost. A separate counter-example keeps the rule itself honest rather than absolute: in Mexico versus England, staying at the market on England’s own 5-or-more-SOT question cost −16.79 RBP relative to what a personal-history-grounded empirical estimate would have won, since the empirical read in that case reflected a still-relevant behavioural pattern rather than a thin, untested sample. The honest generalisation across all three matches is narrower than “trust the market over your own SOT projection”: the market specifically beats an upward extrapolation from a thin team-level sample toward a count threshold, and specifically loses when the deviation reflects a genuine, regime-tested personal pattern rather than an optimistic team-level projection.

Independent per-team decomposition for compound questions. For any “both sides do X” question with no direct market, I now default to modelling each side’s own rate independently rather than assuming a shared or symmetric prior. The clearest validation is Brazil versus Norway’s “will both teams receive at least one card” question: Brazil’s own four-game card rate ($\lambda = 1.75$) and Norway’s own ($\lambda = 0.5$) combine to $P(\text{both carded}) = (1 - e^{-1.75})(1 - e^{-0.5}) = 0.325$, submitted at 0.33 against a crowd of 0.62; the outcome was NO, for +59.43 RBP, the best single result in the campaign to date. The crowd’s 62% reads as a generic high-stakes-knockout heuristic applied without checking either team’s actual disciplinary record; pricing off each team’s own measured rate instead captured a gap a shared prior could not see.

The pattern across nearly every rule above is consistent: initial confirmation at small n , followed by a named failure that forces a more conditional, context-sensitive formulation.

Rules confirmed at $n = 1-3$ are explicitly designated “watch items” rather than settled doctrine.

6 Rejection of Learned Blending Models

The pipeline in §4 is a portfolio of hand-derived, individually theory-grounded rules—an Elo update, a Poisson mean, a named situational adjustment for a specific player’s documented drought. That is exactly the kind of estimator a learned model is supposed to make obsolete: fit one flexible function to the historical record of my own estimates, the crowd’s estimates, and the resulting outcomes, and let the data discover the blending weights rather than hand-tuning them match by match. I built and validated that model, more than once, before rejecting it, and I report the rejection in full because it is the reason the rest of this paper does not contain a learned blend anywhere.

6.1 Four Independent Tests

Platt-scaling recalibration. A two-parameter logistic fit,

$$\text{logit}(P(\text{outcome} = 1)) = a + b \times \text{logit}(\hat{p}_{\text{submitted}}) \quad (3)$$

run on the live submission record, produces global estimates $\hat{a} = -0.21$, $\hat{b} = 0.51$ —nominally suggesting overconfidence—but the bootstrap 95% confidence interval for b comfortably includes 1.0 (perfect calibration), and a walk-forward check shows the fitted slope is itself unstable across the split ($b = 0.196$ out-of-sample versus 0.510 in-sample) while applying the correction degrades out-of-sample Brier by 0.028.

A learned blend of my estimate and the crowd’s. On a filtered sample of 404 group-stage questions across 42 matches, I fit nineteen candidate features (the two raw probabilities and their gap, Elo differential, rest-days differential, squad-value differential, FIFA-ranking differential, pre-match draw probability, and nine binary indicators for named rules that fired at least ten times in-sample) into two candidate models: an L_2 -penalised logistic regression and a shallow histogram gradient-boosting classifier. Three zero-parameter references—trusting the crowd alone, trusting my own estimate alone, and a naïve 50/50 average—set the bar either model had to clear. Table 6 reports both a time-ordered walk-forward validation and a six-fold grouped cross-validation (grouped by match, so no single match’s questions split across the train/test boundary), against the strongest baseline available on identical held-out rows.

Table 6: Learned-model performance against the crowd baseline, two validation schemes.

Scheme	n_{test}	Model	Brier	Best baseline	RBP-equiv.
Walk-forward	208	Logistic regression	0.2545	0.2157	−817.6
Walk-forward	208	Gradient boosting	0.2227	0.2157	−147.9
Grouped 6-fold	404	Logistic regression	0.2569	0.2139	−1,760.3
Grouped 6-fold	404	Gradient boosting	0.2315	0.2139	−721.1

Both models, under both schemes, are worse than simply trusting the crowd, by a margin equivalent to hundreds to nearly two thousand relative Brier points over the test window—an order of magnitude larger than any single-match result in this campaign. Feature-importance diagnostics on the full sample show every one of the nine named-rule indicators contributes essentially nothing beyond the continuous features already in the model: permutation importances are dominated by `crowd_estimate` and `fifa_ranking_diff`, with every rule flag near zero. As Harvey et al. (2016) would predict, this does not mean the rules lack genuine out-of-sample edge, only that a learned model cannot yet separate that edge from noise at this sample size. A proportionality constant fit directly against 386 previously scored questions with a known realised RBP gives $\hat{S} \approx 101.3$ RBP per unit of Brier improvement, the conversion used throughout this section.

A linear regression of RBP against crowd-deviation. Run on a broader, less-filtered sample of 771 rows across 71 matches, a regression of realised RBP on the gap between my estimate and the crowd’s gives an in-sample R^2 of only 0.053, with only a player-prop indicator variable reaching significance; the same model loses to a naïve baseline out-of-sample under grouped cross-validation (mean-squared error 157.23 versus a baseline of 149.59). A small companion exercise on a single player’s own scoring history, illustrative rather than deployed, makes the mechanism visible: an in-sample R^2 of 0.431 on nine observations looks strong until the corresponding F-test fails to reach significance ($p = 0.677$) and leave-one-out cross-validation shows the model losing to the player’s own naïve career average—exactly the overfitting pattern the events-per-variable literature predicts at this sample size (Peduzzi et al., 1996; Riley et al., 2019).

A closed-form result on the cost of deliberate deviation. Because RBP is an addi-

tive, per-question, strictly proper scoring rule rather than a convex or rank-based payoff, deliberately pushing a submission a distance δ past a genuinely calibrated estimate has an expected cost of exactly $S \cdot \delta^2$ under the scale constant above, a guaranteed loss with no offsetting expected benefit; the Kelly-criterion intuition that justifies aggressive position-sizing under a multiplicative, compounding payoff does not transfer to an additive, bounded scoring rule, and the correct frame is closer to a tracking-error budget than a betting-edge calculation. This is not a purely theoretical caveat: the one live attempt in this campaign to apply a deliberate, theory-motivated deviation past a calibrated market estimate (§5, Canada versus Morocco) cost -80.83 RBP, the campaign’s worst single result, for exactly the reason the theory predicts—the underlying signal was not in fact high-confidence, and the quadratic cost of being wrong about that outweighed any plausible benefit.

6.2 Interpretation

Beating the crowd-alone baseline requires extracting real signal from as many as nineteen features using two to four hundred rows clustered into a few dozen truly independent matches. Peduzzi et al. (1996) and its extension in Riley et al. (2019) put the safe threshold at 10–20 events per fitted parameter, capping a defensible feature count at this sample size well below what either candidate model used, and Mamlouk et al. (2024)’s finding that calibration-based model selection reliably beats accuracy-based selection in sports betting is the reason Brier score, not classification accuracy, is the only metric used throughout this section. This is not an implementation failure to be fixed with more careful tuning; it is what “not enough independent data yet” looks like when measured with an honest out-of-sample test rather than an in-sample fit. I deploy neither learned model, and I treat the hand-built probabilistic pipeline of §4—individually theory-grounded, individually falsifiable, and each rule traceable to a specific won or lost question—as the higher-information estimator until the settled record is an order of magnitude larger than it is today. Every probability reported anywhere in this paper follows from that decision.

7 Calibration and Validation

7.1 Historical-Panel Calibration (49,000-Match Backtest)

The walk-forward backtest of the goals-based pipeline surfaces a persistent, approximately uniform 5–8 pp compression gap: at every predicted $P(\text{home win})$ decile from 0.1 to 0.9, the actual home-win rate exceeds the prediction by a broadly similar margin throughout the range. Re-fitting the Dixon–Coles ρ parameter produced negligible movement, ruling out the low-score correction as the primary driver.

The leading hypothesis, documented in `data/calibration_research_notes.md`, is attenuation bias from the Elo covariate: because Elo is itself a noisy, partially bootstrapped estimate of true team strength, its direct inclusion in a regression attenuates the fitted Elo-difference coefficient toward zero, which mechanically compresses predicted probabilities toward 50/50, precisely the observed pattern. Glicko and TrueSkill’s explicit treatment of rating uncertainty as a structural quantity, rather than a point estimate, is a principled alternative I have not implemented. The recommended next diagnostic step, not yet confirmed run, is to re-plot the calibration gap in logit space and separately for neutral versus non-neutral matches, since that split alone discriminates between the attenuation-bias hypothesis and a home-field-advantage miscalibration hypothesis.

7.2 Live-Record Calibration

The Platt-scaling diagnostic reported in full as the first of four tests in §6 is, at this sample size, statistically inconclusive rather than a confirmed finding of overconfidence, and I apply no calibration correction until the live record grows substantially larger. One structural finding does survive this caution: questions where my submitted estimate falls in the 70–100% range earn a mean RBP of only +0.65, compared with +2.7 to +3.1 in every other estimate band, consistent with the prediction that when both I and the crowd are already confident, the exploitable gap between us is minimal.

8 Track Record and Results

8.1 Headline Numbers and a Discrepancy Note

Multiple project documents report cumulative RBP totals at different points in the campaign’s timeline, and I report the disagreement rather than pick one silently.

Table 7: Cumulative RBP reported across project documents (different snapshots)

Source	Matches	Questions	RBP	As of
master_question_dataset.csv (most recent rebuild)	85	921	+3,246.04	Most recent rebuild
France vs. Morocco (verified, not yet merged)	+1	+14	+169.17	2026-07-09
Combined, current	86	935	$\approx$+3,415.21	This writing
Earlier record	49	436	+872.13	2026-07-01

The master-dataset figure is treated as authoritative for everything already merged, since it is derived directly from raw per-match JSON files with an explicit cross-validation step (MASTER_DATASET_DICTIONARY.md independently confirms a dozen RBP figures against the dataset, all matching exactly). France versus Morocco, the most recent settled match at time of writing, is reported separately because the next dataset rebuild has not yet run; I do not fold it into the CSV-derived total by hand. Beat-crowd rate across the 921 scored questions in the current rebuild is 69.7% (642 of 921), up from 58.2% reported at the July 1 snapshot—an improvement worth treating cautiously rather than as a clean trend, since it reflects both genuine rule maturation and a still-accumulating, still-noisy sample.

8.2 Individual Match Results

8.3 Two Matches in Detail

Brazil versus Norway (round of 16, July 5) is documented in full elsewhere in this project and summarised here because it validates two methodological findings at once. Norway entered as a 20% pre-match underdog and won 2–1, and eight of nine pure direct-market questions beat the crowd by five to fourteen RBP each, the unglamorous backbone that keeps a match profitable independent of any single standout call. The match’s two largest

Table 8: Selected individual match results (not an exhaustive listing of all 86 settled matches)

Match	Score	RBP	Beat Crowd	Key result
BRA-NOR (2026-07-05)	Brazil 1-2 Norway	+179.26	13/15	Both-teams-carded compound: +59.43
FRA-MAR (2026-07-09)	France 2-0 Morocco	+169.17	13/14	Priced with zero market anchor at open
MEX-ECU (2026-06-30)	Mexico 2-0	+154.20	11/15	Q12 offsides +49.58
FRA-SWE (2026-06-30)	France 3-0	+148.01	14/15	Direct markets dominate
NED-MAR (2026-06-30)	1-1 (NED out)	+120.30	12/15	Brobbey SOT +24.84
ENG-CDR (2026-07-01)	England 2-1	+119.36	11/15	ENG 20+ shots +27.51
CAN-MAR (2026-07-04)	Canada 0-3 Morocco	+58.92	12/15	Worst single result: Canada 4+ SOT -80.83
JPN-SWE (2026-06-22)	1-1	+70.49	8/10	SWE more SOT +23.98
BEL-EGY (2026-06-21)	1-1 (BEL win)	+63.72	9/10	SOT differential
GER-CUR (2026-06-14)	Germany 7-1	+72.47	9/10	Underdog props x5
CAN-BIH (2026-06-12)	1-1	+42.65	8/10	BTTS props

wins came from the independent per-team decomposition rule (§5, +59.43 RBP) and from correctly differentiating two superficially identical “hot streak” player props by whether the streak had been tested against comparable opposition rather than by raw hit rate alone.

France versus Morocco (quarterfinal, July 9) is the only match in this campaign priced with no crowd consensus or sharp market available for any of its fifteen questions at the moment pricing began, the first time in forty-nine prior matches that neither anchor existed. All fifteen questions were therefore priced by the probabilistic pipeline of §4 alone; a Smarkets quote became available only after the initial pass, for twelve of the fifteen questions, and was used to revise rather than originate the estimates. The match settled France 2–0 Morocco for +169.17 RBP across fourteen submitted questions (one question was deliberately withdrawn rather than submitted on a manufactured justification, discussed in the standalone case study), a new campaign best at the time it settled.

8.4 Pricing-Technique Category Audit

Across the twenty matches in the dataset carrying an explicit pricing-technique tag—question-category coverage is sparse in the raw files, so this is a representative sample rather than the full campaign, and I harmonise two category vocabularies the raw files use across different points in the project’s history rather than treat them as separate schemes—210 questions classify cleanly into seven techniques (Table 9).

Table 9: Performance by pricing technique, harmonised sample (210 questions, 20 matches)

Category	N	Total RBP	Avg RBP	Beat-Crowd
DIRECT (market matches question ~1:1)	99	+365.23	+3.69	69.7%
TEAM_MODEL (Poisson/Skellam from team markets)	65	+234.03	+3.60	72.3%
PLAYER_LIQUID (bid+offer player market, near-raw)	20	+209.74	+10.49	85.0%
PLAYER_ILLIQUID (offer-only, $\times 0.945$)	16	+116.53	+7.28	68.8%
BASE_RATE	4	+25.23	+6.31	75.0%
TIMING_EARLY (before 1st hydration break)	5	+17.21	+3.44	40.0%
TIMING_NO_MARKET (self-built late-window)	1	+11.76	+11.76	100%
Total	210	+979.73	+4.67	71.4%

Two changes from an earlier, smaller audit of this same taxonomy are worth stating directly rather than updating silently. **PLAYER_ILLIQUID** was net negative (-14.27 total RBP) in a four-match, sixty-question snapshot taken early in the knockout stage; it is net positive here, consistent with the 5.5% illiquid-market discount rule maturing since that snapshot. **TIMING_NO_MARKET**, that earlier snapshot’s one structurally loss-making category (-63.72 total, -9.10 average RBP, 42.9% beat-crowd), has nearly disappeared from the mix here, appearing in only one of 210 questions—the direct, intended consequence of a standing rule adopted specifically because of that finding: shade or avoid late-window, no-market questions rather than continue self-building an estimate for them. Both changes are evidence a named rule actually changed behaviour and the behaviour change actually paid off, not just a bigger sample telling a different story by chance.

8.5 Best and Worst Individual Results

The best single-question result in the campaign is Brazil versus Norway’s both-teams-carded compound question, +59.43 RBP, discussed in §5. The best single-match result is Brazil versus Norway overall, +179.26 RBP.

The worst single-question result in the campaign is Canada versus Morocco’s “Canada 4 or more shots on target” question, −80.83 RBP, discussed in full in §5 and §6. I report this in place of an earlier, smaller worst-result citation (a −51.97 RBP personal-conviction override in an early group-stage match) because the newer result is both larger in magnitude and a more useful failure to have on record: it is a traceable methodological failure mode that had already been named and warned about in writing before it happened again, not a one-off lapse in process discipline. The earlier result remains the motivation for the standing rule against ever submitting outside roughly the 0.05–0.95 range, and both losses are kept in this paper rather than superseding one another, since they illustrate two different failure modes worth guarding against separately.

8.6 External Validation: Leaderboard Standing

A single match’s RBP, even a campaign-best one, is a noisy statistic on its own. A separate and largely independent check is the platform’s global leaderboard, aggregating every participant’s cumulative score across every question submitted all tournament. Three snapshots, captured over the course of the tournament, are reported in Table 10.

Table 10: Global leaderboard standing over time (field size and rank independently verified from platform snapshots; July 10 figures self-reported at time of writing).

Date	Rank	Field size	Percentile	Cumulative score
2026-06-27	200	3,564	Top 5.61%	1,059.56
2026-06-29	177	3,628	Top 4.88%	1,245.36
2026-07-10	43	3,879	Top 1.11%	3,568.30

The field itself grew over the window (3,564 to 3,879, +8.8%), so percentile rather than raw rank is the comparable statistic across snapshots: standing moved from the top 5.6% to the top 1.1% of an open, global field over roughly two weeks spanning the transition from

the group stage into the knockout rounds. Cumulative score more than tripled over the same window, consistent with, though not solely attributable to, the knockout-stage results reported across this campaign's individual matches. I do not have a verified breakdown of the competing field's composition and make no claim about it; what is independently verifiable is the field size and the rank within it, and by that measure the standing reported here places this methodology in the top percentile of an open field with no entry barrier and several thousand active participants, moving in the same direction the match-level and question-level results above already point.

9 Limitations and Threats to Validity

Live-record calibration remains genuinely open. As detailed in §7, the Platt-scaling diagnostic cannot statistically distinguish genuine overconfidence from sampling noise at the current sample size. I do not currently know with confidence whether my live estimates are systematically miscalibrated; if they are, the effect is neither large enough nor stable enough to justify a correction yet.

The precise functional form of the JTC’s Relative Brier Points metric has not been independently confirmed against the live platform. Whether the platform’s scoring behaves more like plain Brier, which does not respect the ordinal Home/Draw/Away structure, or like RPS, which does, carries strategic implications: under RPS, hedging probability mass toward Draw on direction-uncertain match-winner questions is specifically rewarded, and under plain Brier it is not. This has not been tested empirically against JTC’s actual formula.

The Elo data defect documented in §4 remains only partially resolved. The point-in-time replay script corrects individual matches on demand, but the underlying source file, `results.csv`, still records every 2026 result as missing, so the defect will recur on every future in-progress fixture until the pipeline is fixed at its source rather than patched match by match.

Three features of the World Cup forecasting setting are not fully captured by any model in this stack. Teams play at most three group-stage matches before elimination becomes possible, so the three-match ESPN in-tournament sample and the 128-match StatsBomb historical panel constitute the entirety of the evidence available for any given team, unlike the multi-season club-level panels the Dixon-Coles and Karlis-Ntzoufras lineages were built for. International teams also play infrequently outside major tournaments and squad composition changes substantially between World Cups, so a four-year-old Elo rating carries considerably less information than a four-year-old club rating drawn from a continuous domestic competition would. Finally, a documented failure mode recurs across multiple matches: applying a team-archetype lesson learned from one team to a different team that shares only superficial characteristics, as happened when an “elite teams produce low shot volume” pattern learned from one South American side was incorrectly transferred to a different team entirely, producing six below-crowd misses on attacking-volume questions in

a single match.

This remains single-investigator research conducted under competitive time constraints and evaluated against a partially opaque scoring mechanism, without peer review. All rules in their present form should be regarded as provisional, in particular those confirmed on samples of $n = 1-3$ observations, and §5's own Canada-Morocco result is a reminder that a rule validated once can still fail badly the next time it is applied to a signal that only superficially resembles the one that validated it.

10 Future Work

The most consequential near-term improvement is procedural rather than statistical. A Kane transcription error, a missed submission deadline, and a corner-mapping near-miss together account for more than 140 RBP of avoidable loss across the campaign, with no modelling disagreement in any instance. A final value-by-value comparison between what is about to be submitted and the researched recommendation, before confirming, would eliminate this class of error entirely. No model improvement identified in this paper approaches that magnitude of recoverable loss.

Several diagnostics remain genuinely outstanding rather than merely unfinished. Merging France versus Morocco, and any subsequent semifinal or final results, into the master dataset rebuild is the most immediate item, since every headline number in this paper currently carries a small, explicitly flagged gap while that match sits outside the rebuilt CSV. Re-running the neutral-versus-non-neutral backtest split in logit space would confirm whether the Poisson pipeline’s residual calibration gap originates in the now-corrected train/predict mismatch or in errors-in-variables attenuation that would persist regardless. Extending the point-in-time Elo replay beyond the five retrospective matches already covered, to every remaining knockout fixture, would close the gap documented in §4 properly rather than match by match. The live Platt-scaling record has a natural checkpoint once the submission count grows substantially past its current size, at which point the bootstrap confidence interval on b in Equation~(3) may finally exclude $b = 1.0$ with enough precision to justify a correction. A per-market-type gradient-boosted model trained on the StatsBomb panel, already downloaded and flattened into team- and player-match panels, is a concrete next step for fouls, cards, shots-on-target, corners, and offsides base rates specifically, validated with the same walk-forward and grouped-by-match discipline that rejected the crowd-blending models in §6—a narrower, better-specified modelling target than “blend my estimate with the crowd’s” might clear the events-per-variable bar this paper’s own rejection argument currently fails.

Finally, simulating the five canonical scenarios from Constantinou and Fenton (2012) against the platform’s revealed scores would establish empirically whether JTC’s formula behaves like Brier or RPS, with direct implications for the hedging strategy on three-way match-winner questions, and stratifying the live record by question family—score propositions,

goal-total propositions, shots on target, timing—would test whether the miscalibration identified by the global Platt diagnostic concentrates in specific categories rather than sitting uniformly across all of them.

11 Conclusion

This paper documents a solo forecasting campaign for the 2026 FIFA World Cup, built on a formal statistical pipeline and a systematic literature review, and evaluated against a live, proper-scoring-rule mechanism across 935 individually resolved questions.

Two findings anchor it, and they are worth holding together rather than reading as separate results. The first is operational: the JTC crowd consensus is systematically compressed toward 50% relative to a calibrated probability, a relationship that held essentially unchanged when the sample doubled from the group stage into the knockout rounds, and this compression, not any model’s absolute accuracy advantage, is the dominant source of positive RBP. The second is structural and, at this campaign’s sample size, decisive: four independent tests, run at different points and different sample sizes, all reach the conclusion that a learned blend of my own estimate and the crowd’s is not trustworthy, and a live attempt to override that conclusion once, under a theoretically motivated argument for deliberate deviation, produced the campaign’s single worst result at almost exactly the cost the underlying theory predicts. Neither finding is a statement that learned models are wrong in general; both are quantified statements about where the events-per-variable constraint actually binds for this specific problem at this specific sample size, and both are the reason every probability in this paper came from a hand-built, individually falsifiable rule rather than a fitted blend.

The pricing-technique audit gives a concrete, evidence-graded account of which techniques generate positive expected value: direct market trust and team-level decomposition are reliably profitable, and a previously identified loss-making category, self-built late-window timing with no market reference, has nearly vanished from the mix as the standing rule built to avoid it actually changed behaviour. The leaderboard trajectory—from the top 5.6% to the top 1.1% of an open field of nearly 3,900 participants over three weeks—is the most externally verifiable evidence this paper offers that these findings, in aggregate, translate into a real result rather than a favourable read of my own bookkeeping.

12 Appendix: Data Dictionary Summary

12.1 master_question_dataset.csv Column Inventory (Selected)

Full documentation in `datasets/MASTER_DATASET_DICTIONARY.md`. Built by `datasets/build_master_dataset.py` from raw per-match files in `data/external_markets/` and `matches/`. 943 rows, 107 columns, 85 matches at the most recent rebuild.

Table 11: master_question_dataset.csv: key columns

Column	Description
<i>Match-level (repeated across each match's rows)</i>	
match	Canonical team-code pair, e.g. ARG-AUT
date	Match date
schema_era	1, 2, or 3 (generation of raw-file schema; each handled separately)
final_score	Reported final score
match_total_rbp	Summed RBP across all questions for this match
elo_team_a_pre, elo_team_b_pre, elo_diff	Pre-match Elo ratings and difference
rest_days_a, rest_days_b	Days since each team's previous match
squad_value_a_eur, squad_value_b_eur	Transfermarkt squad values (EUR; null for 5 teams)
draw_probability	Pre-match three-way draw probability from verified market
rule1_fired ...rule15_fired	Boolean flags for each named rule application
<i>Question-level (one value per row)</i>	
question_text	Platform question wording
question_category	Pricing technique: DIRECT, TEAM_MODEL, PLAYER_LIQUID, PLAYER_ILLIQUID, etc.; sparse coverage (210 of 943 rows)
recommended_estimate	Pre-match research figure; matched by topic-word Jaccard similarity (≥ 0.35)
our_estimate	Final submitted probability

Continued on next page

Table 11 continued

Column	Description
crowd_estimate	Platform-revealed crowd consensus after resolution
outcome	Binary outcome (1 = event occurred)
rbp	Relative Brier Points earned (blank, not zero, for unsubmitted questions)
beat_crowd	Boolean: did the submitted estimate outperform the crowd?
submission_error_flag	True where $ p_{\text{submitted}} - p_{\text{recommended}} > 0.03$
actually_submitted	False for questions never reaching the platform before a deadline, or belonging to a calibration-only match
postmortem_note	Free-text root-cause annotation

12.2 Other Datasets Referenced in This Paper

Table 12: Supporting datasets

File	Rows	Cols	Role
data/processed/elo_match_panel.csv	~49,400	—	Point-in-time Elo teams, one row per cal international match (present)
data/international_results/results.csv	49,473	9	Source match results (martj42); foundation statistical modelling every 2026 World Cup missing (§4)
StatsBomb WC2018+WC2022 event panel	6,131 (player-match) / 257 (team-match)	—	Validated historical panel, 128 matches
ml/feature_matrix.csv	264	11	Meta-model input t 06-11 to 2026-06-30

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